

The Origin of Nondestructive Determination of Characteristic Material Parameters Using Electromagnetic Methods

Friedrich Förster, Reutlingen

Conversion of characteristic material parameters to measurable variables using suitable sensors is a field of activity to which I have devoted 57 years of my life. In 1931, I developed a method of converting the growth rate of plants to an acoustic frequency variation at the University of Göttingen. This made it possible to "hear the grass grow" in the true sense of the word. My doctoral thesis which I completed in 1932 dealt with what was then a new ultrasonic method for converting the velocity of sound in fluids to electrical measured values, thus permitting the effect of molecular association on the velocity of sound to be measured.

In 1935 I joined the Kaiser-Wilhelm-Institut of Metallurgical Research in Stuttgart which had then just been founded (and which is now the Max-Planck-Institut). As early as the first few weeks, I realized that the capabilities of electronics, a field which was still entirely new at the time and a discipline in which I, as a physicist, was particularly interested, were still largely unknown in this institute which had just been opened. By contrast, when this research institute was started, there was an extraordinarily great need for new measuring methods to determine structural parameters and their gradients as a function of time and temperature etc.

Owing to my great love of music, I first applied myself to the sound of metallic materials, i.e. the elastic and anelastic properties, and in 1935 I developed a method of measuring the modulus of elasticity and attenuation as a function of temperature and time. For this purpose, it was necessary for me to develop magnetic pickups and sensors

since suitable ones were not available commercially at the time. This acoustic measuring method which led to the adoption of the particularly important characteristic material parameter of attenuation in the field of metallurgical research is now used widely in the fields of metallurgical, ceramic, glass, plastics and wood research, in addition to the corresponding industries.

As early as 1936, I used the acoustic sensor which I developed in 1935 for the initial experimental investigations and recordings of acoustic emission in metallographic processes such as martensite conversion, the conversion of zirconium and the twinning of mechanically stressed bismuth monocrystalline wires. In the following year, 1937, the acoustic sensor which I developed in 1935 was used again for the far-removed field of magnetic measurements and surveys. For this purpose, the ferromagnetic body under investigation was connected to the acoustic sensor by a miniature quartz tube to a very slight extent, e.g. with a weight of a fraction of a gram. The test piece which was located in a small electric furnace was magnetized by means of a DC field coil outside the furnace so that it was characterized by a magnetic moment. This magnetic moment was, at the same time, influenced by the gradient of an alternating magnetic field generated outside the furnace. This subjected the test piece to an alternating force which the acoustic sensor converted to an AC voltage. After amplification, the magnetic moment of the test piece was indicated by a measuring instrument. The measuring instrument could be used to trace physical metallographic processes, such as processes of conversion etc. as a function of temperature and time, through to determining the temperature of the Curie point. In the 30s, this acoustic sensor which was originally developed for totally different purposes and which was patented in the 1937 for magnetic measurements, was used for a whole series of theses.

Even these initial sensors which I developed in 1935 for acoustic measurements taught me something which has been confirmed time and time throughout the long course of my professional career, namely that developing a sensor generally leads to applications which far exceed the initial development objective of the sensor in question.

As early as 1936 to 1937, the manifold problems which face a metallurgical research institute made me realize that electromagnetic methods embodied extraordinary potential capabilities for research and industry. Owing to my lack of experience at the time, I naively assumed that it would be possible to open up the capabilities of using magnetic methods for non-destructive determination of characteristic material parameters simply by conducting series of experimental tests. I asked myself what was the influence of electrical conductivity, magnetic permeability and the dimensions (diameter, tube wall thickness and plating etc.) of metallic test pieces located in the electromagnetic field of a test coil on the impedance of the test coil if an alternating current of frequency f is passed through this coil. I tried to answer this question by conducting extensive series of experimental tests. But this turned out to be a mistake since there is an infinite number of possible test piece parameter and test frequency combinations. Owing to these unsatisfactory, experimental results, I resolved to thoroughly and painstakingly study the entire field of non-destructive determination of characteristic material parameters using electromagnetic methods in order to derive from this the laws of similarity which permit the many and varied influences of the specified parameters to be reduced to one single standardized curve. I was certainly aware that theoretically establishing the effect of influencing parameters of regularly cylindrical bars, tubes, level surfaces and spheres on the electromagnetic data of a through-type coil and the influencing parameters of test pieces and foils on the electromagnetic data of an approach-type coil or a transmission coil array would be no easy task.

For years, I was confronted by many unsolved problems since the extensive experimental results which I had obtained previously could not be incorporated significantly in a generally applicable law before establishing the theoretical background, i.e. the laws of similarity.

However, when the theoretical fundamentals of eddy-current testing were elaborated in 1940, the great understanding of the inner relationships which this provided made it possible to elaborate the endeavored simplification. The infinite number of possible material property combinations: electrical conductivity, magnetic permeability and geometrical dimensions and shapes of the test piece with the frequency of the test instrument could be represented with a normal curve for specific test piece shapes. At the same time, it was possible to theoretically derive the result that variations in the electrical conductivity, permeability and dimensions of a test piece must occur in different directions in the impedance plane of the test coil. This finding was a tremendous challenge involving displaying the impedance plane of the test coil which had then been defined only in theory or displaying the equivalent complex voltage plane of the test coil on the screen of a cathode ray tube in order to draw conclusions in respect of the cause in the test piece on the basis of shifts in the impedance value of the test coil. The method for quantitative display of the impedance plane of the test coil on the screen of a CRT was developed in December 1940.

Nowadays, displaying the impedance plane or the equivalent complex voltage plane of the test coil on the screen of a CRT is a foregone conclusion world-wide. This is why, today, no one could comprehend the excitement and emotion which I felt when I pushed a bar made of an aluminium alloy with a slight local diameter reduction produced by emery paper and local punctiform heating, i.e. a reduction in the electrical conductivity, through the test coil during my first attempt to display the test coil impedance plane on a fluorescent

screen. Before my very eyes, as the section of the bar with reduced diameter passed through the coil, the bright luminous spot on the screen moved in a direction different to the direction in which it moved when the point with the conductivity change passed through the coil, and this was exactly as I had predicated in my theory which I had just concluded. This phenomenon which excited me so much, i.e. that the luminous spot on the screen moved in different directions, just as my theory had predicted, made me realize immediately that this two-dimensional display of the test coil value would become the standard method of eddy-current testing in the future. And today, 48 years later, this eddy current impedance plane analysis method actually is used throughout the world.

Between the early 40s and early 50s, the impedance values or the complex voltage values of the test coil were calculated for the following test pieces:

- 1.) Regularly cylindrical bars made of non-ferromagnetic material for various test coil space factors.
- 2.) Regularly cylindrical bars made of ferromagnetic material for various permeabilities.
- 3.) Thin-walled tubes made of non-ferromagnetic material.
- 4.) Non-ferromagnetic tubes with various wall thicknesses.
- 5.) Non-ferromagnetic bars or thick-walled tubes with an outer plating comprising a non-ferromagnetic metal and characterized by differing electrical conductivity.
- 6.) Non-ferromagnetic thick-walled tubes, with the test coil inside the tube.
- 7.) Ferromagnetic spheres with a high magnetic permeability in a short test coil.
- 8.) Ferromagnetic spheres or short, regularly cylindrical bodies with various magnetic permeabilities in a short test coil.
- 9.) Non-ferromagnetic spheres in a short test coil.
- 10.) Flat sheet material or foil with primary or secondary coil on opposite sides (transmission method).

So-called sensitivity diagrams were calculated for the above test pieces and test coil arrays with the purpose of quantitative evaluation of eddy-current measurements and selecting an optimum frequency for specific tasks involving non-destructive determination of characteristic material parameters. These diagrams show the amplitude and phase angle on the screen when the parameters of property for the various test pieces, such as electrical conductivity, magnetic permeability, diameter and wall thickness, change by 1 %, as a function of the ratio of the eddy current test instrument's test frequency to the fundamental characteristic parameter of the eddy-current theory, namely the limit frequency of the test piece. The appropriate formulae are available for the limit frequencies of the above-listed test pieces with the relevant dimensions and physical properties, termed "Foerster limit frequency" in American literature. All above-mentioned impedance diagrams and the sensitivity diagrams just mentioned have been calculated, tabulated and recorded for the ratios frequency to limit frequency. For this purpose, approximation equations for regularly cylindrical bodies, for spheres, for thin-walled tubes and for thin, flat sheet materials have been derived for low and high values of these ratios. It is only the ratios of frequency to limit frequency which make it possible to apply the laws of similarity of the eddy-current technique for any parameters electrical conductivity and magnetic permeability in addition to dimensions of a test piece for any frequencies of the coil field to these impedance and sensitivity diagrams. This quantitative explanation of the theoretical fundamentals of the eddy-current method which I have but touched upon above and a major part of which is described in the US Non-destructive Handbook Electromagnetic Testing - Editions I and II turns this approximate indicator method into a non-destructive measurement method. And it was just this aim which I set myself exactly half a century ago. I have not discussed eddy-current crack testing because it is not the subject of this congress. But may I, at this point, mention that extensive mercury model tests on which flaws were

simulated by suitable plastic bodies have permitted the complex test voltage effects to be determined for a broad range of test frequency to limit frequency ratios for a large number of flaw configurations and flaw locations. The laws of similarity of the eddy-current theory also now permit us to predict the influences of material flaws on the complex voltage of the test coil, whereby such influences cannot be calculated strictly mathematically.

However, during my research into the experimental and theoretical fundamentals of the eddy-current method in the late 30s, I was disturbed by the fact that, over and above crack testing, it was possible to non-destructively determine only two material parameters - electrical conductivity and magnetic permeability. I must explain that the metallurgical research institute for which I was working urgently required a method of determining numerous physical and technological characteristic material parameters occurring in conjunction with metallurgical processes non-destructively and, for thermal investigations, wherever possible also without physical contact. The main emphasis was on magnetic measurements for answering questions related to metallography. Ferromagnetic investigations, e.g. the method of punctiform, ballistic recording of an hysteresis loop which was conventional at the time, were extremely time-consuming. When, in 1939, the first CRTs with a screen diameter of 18 cm became available, I devoted myself to the task of developing an instrument for stationary display of the hysteresis loop on a fluorescent screen. In 1938, on the occasion of the Institute's annual conference, it was possible to demonstrate this instrument in numerous example applications. In addition to quantitative measurement of the coercive force, remanence, saturation, initial permeability, Rayleigh's constant and magnetostriction as a function of the temperature up to the Curie point, the differential hysteresis loop method provided a sensitivity for non-destructive determination of various characteristic material parameters which extended way beyond the previous magnetic

measurement methods (e.g. by a factor of 1000). The differential hysteresis loop method consists in arranging two identical test pieces, such as wires or strips etc. which are magnetized in a primary field, in two back-to-back-connected secondary coils so that only the difference in the magnetic properties between the two test pieces is displayed on the screen. If one of the two identical test pieces is modified, either thermally or mechanically, the magnetic effect of such a modification can be amplified, e.g. by a factor of 1000, as compared with the sensitivity of the loop display itself. In addition to being used for normal series of measurements, this instrument was used immediately for establishing the true physical limit, i.e. the start of an irreversible hysteresis loop change of a ferromagnetic wire made of iron or nickel as a function of the tempering temperature of these wires. At a tempering temperature of 900 °C, this physical limit was one power of ten below the values determined mechanically until then. As early as 1940, the German "Zeitschrift für Metallkunde" (metallographic periodical) received two extensive papers elaborated with the aid of this new differential hysteresis loop method, entitled "Magnetic Investigations of Internal Stresses". The following characteristic material parameters were investigated with this new instrument which permitted the magnetic properties of ferromagnetic wires with diameters of as low as 50 μm and at field strengths of $H_{\text{max}} = 10^{-3}$ to $H_{\text{max}} = 2 \times 10^3$ Oe to be determined:

- 1.) Quantitative build-up of internal second-order stresses by stretching in the plastic region on ferrous and nickel wires until the wires break.
- 2.) Quantitative tracing of the process of reduction of internal stresses with increasing tempering temperature.
- 3.) Determining the physical yield point when the first irreversible change in remanence occurs as a function of the tempering temperature.
- 4.) Development of first-order internal stresses when burring a wire.

- 5.) Influence of second-order stretching stresses on the first-order stress state of burred wire.
- 6.) Influence of the tempering temperature on the first-order stress state of a burred wire.

The extreme sensitivity for the vertical magnetization axis on the screen of the ferrograph clearly showed the Barkhausen effect on the screen trace. For this reason, investigations were then conducted into what characteristic material parameters could be derived from the Barkhausen effect. Consequently, a paper entitled "On the question of magnetic fold-over processes in iron and nickel" was concluded in October 1940. This paper dealt with the following investigations:

- 1.) The influence of tempering temperature on the mean range magnitude of Barkhausen effects on cold-burred carbonyl iron wire. The same influence was investigated on carbonyl nickel wire. This established the role of first and second-order stresses on the mean range magnitude.
- 2.) The influence of second-order stresses when stretching carbonyl iron wires on the mean range magnitude of the Barkhausen effect.
- 3.) Initial oscillographic traces of the Barkhausen effect as a function of time, carried out in 1939, provided important insights into the interrelationships between the range volumes of the Barkhausen effect and the material properties.
- 4.) Investigation of the reduction of a large Barkhausen effect complex by stretching the wire with increasing initial field strength.
- 5.) Simultaneous recording of the longitudinal and circular components of the same Barkhausen effect with a dual-trace oscilloscope. This clearly establish the mechanism of the Matteucci effect.

To conclude, I shall discuss a sensor for magnetic DC fields, which resulted from the above magnetic investigations and which, in the same way as the eddy current impedance plane

analysis method, is now in use throughout the world as a Förster probe, ferro-probe, flux gate detector or second harmonic detector. This magnetic sensor which, when it was developed in 1939, was the most sensitive device for measuring magnetic DC fields, with high indicating stability for fields with a strength of even less than one hundred thousandth of the earth's magnetic field formed the basis for a whole series of methods and instruments for non-destructive determination of characteristic material parameters. I shall name but a few by way of example:

- 1.) Instruments for determining the coercive force even on very small test pieces, and which today incorporate an automated measuring sequence.
- 2.) Instruments for susceptibility measurement in the low range on non-magnetic steels, in addition to tracing the stability with mechanical stressing of austenitic steels.
- 3.) Residual field test instruments for sorting mass-produced parts on the basis of hardness, tempering temperature, carbon content and case depth.
- 4.) Instruments for measuring the rolling mill texture on ferromagnetic sheets with the aid of the point-pole method.
- 5.) Instruments for measuring magnetic leakage flux with the normal leakage-flux method or with the magnetography method described by the author in 1950. Förster micro-probes with dimensions 1 x 1 x 1 mm are used for this purpose.
- 6.) Detection instruments with Förster differential probes for quantitative measurement of relative characteristic material parameters of unknown sought objects concealed either underground or on or below the seabed. These characteristic material parameters relate to: magnitude of the magnetic moment, axial direction of the magnetic moment in space, depth of the sought object and point on the surface perpendicularly above the focus of the sought object's center of magnetic moment. Detection devices equipped with Förster differential probes have

been used since the war to locate millions of unexploded bombs and mines, thus permitting them to be disposed of.

- 7.) Finally, these magnetic field sensors surveyed the magnetic field and, thus, the magnetic moment of the planet Venus, as the start of the series of space projects as early as 1962 by the American "Mariner 2" space probe. Here we have a planet as the test piece and the planet's magnetic moment as the characteristic material parameter.